# Shahnawaz A. Adil, Ph.D.

shazadil14@gmail.com
https://shan1525.github.io/
https://inspirehep.net/authors/2726490
ICF-UNAM, Cuernavaca, Mexico
(+91) 9999217668

## Current Position

2024 – Present **Postdoctoral Fellow**, Instituto de Ciencias Físicas (ICF), UNAM, Mexico.
Working on dark energy, cosmological tensions, and large-scale structure
Advisor: Dr. José Alberto Vázquez González

## Research Interests

- **Cosmology**: Dark Energy, Large Scale Structure, Hubble tension, $S_8$ tension, Modified gravity, dark matter–dark energy interaction, scalar fields.

## Education

2018 – 2024 **Ph.D. in Physics**, Jamia Millia Islamia, Delhi, India
Thesis: *Cosmological non-standard field dynamics in light of observations*
Advisor: Dr. Somasri Sen

2014 – 2016 **M.Sc. in Physics**, Jamia Millia Islamia, Delhi, India
CGPA: 8.04/10 (Distinction)
Thesis: *Poincaré–Cartan forms for extended phase space in curved background*
Supervisor: Prof. Pankaj Sharan

2011 – 2014 **B.Sc. (Hons.) Physics**, Jamia Millia Islamia, Delhi, India
First Division (68%)

## Publications

### Journal Articles


1. **S. A. Adil**, M. A. Zapata, Ö. Akarsu, and J. A. Vazquez, "Background-level reconstruction of scalar-field potentials from dark-energy histories and comparison with analytic potential families," Mar. 2026. arXiv: 2603.14693 [astro-ph.CO].

2. D. Kumar, A. Mitra, **S. A. Adil**, and A. A. Sen, "Exploring alternative cosmologies with the LSST: Simulated forecasts and current observational constraints," *Phys. Rev. D*, vol. 111, no. 4, p. 043 503, 2025. DOI: 10.1103/PhysRevD.111.043503. arXiv: 2406.06757 [astro-ph.CO].

3. E. Specogna et al., "Updated Constraints on Omnipotent Dark Energy: A Comprehensive Analysis with CMB and BAO Data," Apr. 2025. arXiv: 2504.17859 [gr-qc].

4. M. A. Zapata, G. Garcia-Arroyo, **S. A. Adil**, and J. A. Vazquez, "How Holographic is the Dark Energy? A Spline Nodal reconstruction approach," Jul. 2025. arXiv: 2507.22292 [astro-ph.CO].

5. **S. A. Adil**, M. G. Dainotti, and A. A. Sen, "Revisiting the concordance ΛCDM model using Gamma-Ray Bursts together with supernovae Ia and Planck data," *JCAP*, vol. 08, p. 015, 2024. DOI: 10.1088/1475-7516/2024/08/015. arXiv: 2405.01452 [astro-ph.HE].

6 **S. A. Adil** et al., "Omnipotent dark energy: A phenomenological answer to the Hubble tension," *Phys. Rev. D*, vol. 109, no. 2, p. 023 527, 2024. DOI: 10.1103/PhysRevD.109.023527. arXiv: 2306.08046 [astro-ph.CO].

7 N. Menci, **S. A. Adil**, U. Mukhopadhyay, A. A. Sen, and S. Vagnozzi, "Negative cosmological constant in the dark energy sector: tests from JWST photometric and spectroscopic observations of high-redshift galaxies," *JCAP*, vol. 07, p. 072, 2024. DOI: 10.1088/1475-7516/2024/07/072. arXiv: 2401.12659 [astro-ph.CO].

8 **S. A. Adil**, U. Mukhopadhyay, A. A. Sen, and S. Vagnozzi, "Dark energy in light of the early JWST observations: case for a negative cosmological constant?" *JCAP*, vol. 10, p. 072, 2023. DOI: 10.1088/1475-7516/2023/10/072. arXiv: 2307.12763 [astro-ph.CO].

9 **S. A. Adil** et al., "S8 increases with effective redshift in ΛCDM cosmology," *Mon. Not. Roy. Astron. Soc.*, vol. 528, no. 1, pp. L20–L26, 2023. DOI: 10.1093/mnrasl/slad165. arXiv: 2303.06928 [astro-ph.CO].

10 A. Bassi, **S. A. Adil**, M. P. Rajvanshi, and A. A. Sen, "Cosmological evolution in bimetric gravity: observational constraints and LSS signatures," *Eur. Phys. J. C*, vol. 83, no. 6, p. 525, 2023. DOI: 10.1140/epjc/s10052-023-11707-4. arXiv: 2301.11000 [astro-ph.CO].

11 Ruchika, **S. A. Adil**, K. Dutta, A. Mukherjee, and A. A. Sen, "Observational constraints on axion(s) dark energy with a cosmological constant," *Phys. Dark Univ.*, vol. 40, p. 101 199, 2023. DOI: 10.1016/j.dark.2023.101199. arXiv: 2005.08813 [astro-ph.CO].

12 A. A. Sen, **S. A. Adil**, and S. Sen, "Do cosmological observations allow a negative Λ?" *Mon. Not. Roy. Astron. Soc.*, vol. 518, no. 1, pp. 1098–1105, 2022. DOI: 10.1093/mnras/stac2796. arXiv: 2112.10641 [astro-ph.CO].

13 **S. A. Adil**, M. R. Gangopadhyay, M. Sami, and M. K. Sharma, "Late-time acceleration due to a generic modification of gravity and the Hubble tension," *Phys. Rev. D*, vol. 104, no. 10, p. 103 534, 2021. DOI: 10.1103/PhysRevD.104.103534. arXiv: 2106.03093 [astro-ph.CO].

## Skills

| | |
|---|---|
| Languages | English (fluent), Spanish (poquito), Hindi, Urdu |
| Coding | Python, Bash, LaTeX |
| Scientific Tools | CLASS, CAMB, MontePython, Cobaya, SIMPLEMC, GetDist |
| Inference Methods | Emcee, MultiNest, PolyChord, George, GaPP |
| Computing | Linux, Windows, HPC environments, MPI parallelization |

## Conferences

| | |
|---|---|
| Nov 2024 | Presented a talk at *IX Essential Cosmology for the Next Generation*, Playa del Carmen, Mexico, on "Dark Energy in the Light of JWST Anomalies". |
| | Presented a colloquium talk at *Instituto de Ciencias Físicas (ICF), UNAM*, on "13.8 billion years in the making: the evolution of our Universe and its unsolved mysteries". |
| Aug 2024 | Presented a talk at *21st Century Cosmology: Tensions, Anomalies & New Physics*, Ashoka University, India, on "Cosmological tensions and new physics in the dark energy sector". |
| Dec 2022 | Presented a talk at *Indian Association for General Relativity and Gravitation (IAGRG)*, on "Is there an evidence of negative cosmological constant through recent cosmological observations?". |
| Nov 2022 | Presented an e-poster at *Young Astronomers' Meet*, ARIES, Nainital, titled "Evidence of an AdS vacua in the Universe". |

## Conferences (continued)

| | |
|---|---|
| Oct 2022 | Presented a talk at *Cosmology@CCSP_1*, on "Late-time acceleration due to a generic modification of gravity and Hubble tension". |
| Nov 2021 | Presented a talk at *Midwest Relativity Meeting*, on "Late-time acceleration due to a generic modification of gravity and Hubble tension". |

## Outreach & Mentorship

| | |
|---|---|
| 2026 | Co-advisor of **Andoni Fontanet Fernández** for the BSc degree project titled, "**Simetrías del espacio-tiempo y transporte del momento angular en gases diluidos relativistas usando Teoría Cinética**". |
| | Co-advisor of **Andrea Valencia Reyes** for the BSc degree project titled, "**Algoritmos para el estudio de la energía oscura**". |
| 2024 | Taught a cosmology program to school children (link). |
| 2023 | Co-mentored Dharmendra Kumar for Ph.D. course work. |
| 2022 | Co-mentored Amresh Verma for M.Sc. dissertation. |
| 2021 | Co-mentored Sonej Alam for Ph.D. course work. |
| | Volunteered for science outreach program for **Square Kilometre Array (SKA)** in Vigyan Samagam. |

## Schools & Seminars

| | |
|---|---|
| Sept 2024 | Regular participant in weekly seminars at **Instituto de Ciencias Físicas (ICF), UNAM**. |
| July 2023 | Attended *Summer School on Cosmology*, ICTP. |
| July 2022 | Attended *Summer School on Cosmology*, ICTP. |
| Dec 2020 | Attended the 31st meeting of **IAGRG**, IIT Gandhinagar. |
| Sept 2020 | Attended *Physics of the Early Universe*, ICTS (online). |
| Nov 2019 | Attended **SERB Preparatory School in Theoretical High Energy Physics (THEP)**, Tezpur University. |

## Teaching Experience

| | |
|---|---|
| Feb 2025 | Teaching Assistant for the course "Machine Learning and Cosmology" (Instructor: Dr. José Alberto Vázquez González). |
| Mar 2023 | Taught physics to NEET/JEE aspirants at Orchid Institute, New Delhi (link). |
| Aug 2021 | Conducted B.Sc. Physics laboratory sessions, Department of Physics. |

## References


**Prof. Somasri Sen**
Jamia Millia Islamia, India
somasri.sen@gmail.com

**Dr. J. Alberto Vázquez González**
ICF-UNAM, Mexico
javazquez@icf.unam.mx

**Prof. Eleonora Di Valentino**
University of Sheffield, UK
eleonora.divalentino@gmail.com

**Prof. Anjan A. Sen**
Jamia Millia Islamia, India
anjsen@gmail.com

**Dr. Sunny Vagnozzi**
University of Trento, Italy
sunny.vagnozzi@unitn.it

**Dr. Özgür Akarsu**
Istanbul Technical University, Turkey
akarsuo@itu.edu.tr